Chapter 1

Introduction to Markov Chains

1.1 Markov Chains

Today was a `Sunny` day. However, I would like to know if it will be `Sunny` tomorrow. These are two categorical states of the weather.

How can I predict the weather? In this case, we are trying to determine values of *uncertainty*. We are uncertain of the weather the next day, or even, the weather in the next 5 days. How many days previously do I need to know the weather to make an accurate prediction?

This is the question that Russian mathematician Andrey Markov curious about:

> How much history of a state do you need to predict a current state?

His discovery was the following: you only need to know the current state to predict the *next state*. This is now known as the *Markov property*. We say a process is *Markovian* if, given the current state, knowledge of past states provides no additional information about the next state. This requires that the future state depends only on the present state and not on the sequence of events that preceded it. In other words, the process has no memory of past states. Generally, this chapter will deal with *discrete-time Markov chains*, where the process evolves in discrete time steps. We will discuss continuous-time Markov chains in a later chapter.

Suppose there are 3 days: Yesterday, Today, and Tomorrow. Information about Yesterday's weather does not provide any additional information about Tomorrow's weather once we know Today's weather.

$$\mathrm{P}(\text{Tomorrow} \mid \text{Today}, \text{Yesterday}) = \mathrm{P}(\text{Tomorrow} \mid \text{Today})$$

While weather, in reality, is not perfectly Markovian, this property allows us to model and predict it using a simplified framework. In a Markov chain, we can represent the states (`Sunny`, `Rainy`, `Cloudy`) and the probabilities of transitioning from one state to another.

Suppose we classify the weather as either `Sunny` or `Rainy`. We would call the *state space* $\mathcal{S} = \{\texttt{Sunny}, \texttt{Rainy}\}$. Notice the special S in the notation $\mathcal{S}$, which stands for *state space*.

Based on historical observations, we might estimate the following probabilities:

- If today is `Sunny`, tomorrow will be `Sunny` with probability 0.8 and `Rainy` with probability 0.2.
- If today is `Rainy`, tomorrow will be `Sunny` with probability 0.4 and `Rainy` with probability 0.6.

These probabilities completely describe our Markov chain. Once we know today's weather, we can predict the probabilities for tomorrow's weather without needing to know what happened last week.

The states and transitions probabilities can be represented using a *Markov chain*, also called a *state diagram*,(which looks similar to a graph!):

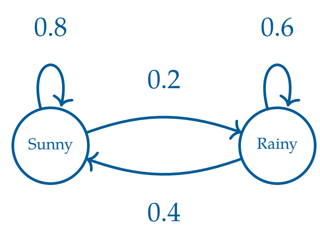

Figure 1.1: A Markov chain representing the weather with two states: `Sunny` and `Rainy`. The arrows indicate the probabilities of transitioning from one state to another.

Here, given the state of the weather today, we can predict the probabilities of the weather tomorrow by following the chain of arrows. There is a great visualization of Markov chains in your digital resources.

We can generalize the *Markov property* as the following equation:

$$P(X_{n+1} = j \mid X_0, X_1, \dots, X_n) = P(X_{n+1} = j \mid X_n)$$

where $X_0, X_1, \dots, X_n$ are states of a given process at times $0, 1, \dots, n$ respectively. This means that the probability of transitioning to a next or future state j depends only on the current state X_n and not on any of the previous states.

As the number of states increases, the Markov chain can become more complex. However, the Markov property still holds, allowing us to model and analyze processes in the form of a matrix of transition probabilities. This matrix, known as the *transition matrix*, captures the probabilities of moving from one state to another in a compact form.

Exercise 1 Frank's Hot Dog Chain

Working Space

Frank is opening a new hot dog stand. Depending on how much they like his hot dogs, new customers have a chance of becoming a regular customer. If they do not like them, they may become an inactive customer.

A new customer has a:

- 40% chance of remaining new,
- 50% chance of becoming regular,
- 10% chance of becoming inactive.

A regular customer has a:

- 10% chance of reverting to new,
- 80% chance of staying regular,
- 10% chance of becoming inactive.

An inactive customer has a:

- 20% chance of returning as new,
- 30% chance of becoming regular,
- 50% chance of remaining inactive.

Assuming the X_n are the states at of the n-th visit to the store (such that $X_0 \implies$ First Visit),

1. Identify the state space of this Markov chain.
2. Draw a Markov chain to represent this process.
3. Simplify this using the Markov property, and solve for the following probability:

   $$P(X_{n+1} = R \mid X_n = I, X_{n-1} = R, X_{n-2} = N)$$

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1.2 Transition Matrices

A *transition matrix* (also sometimes called a *stochastic matrix*) records the probability of moving from one state to another in a Markov chain. Each row corresponds to the *current state*, and each column corresponds to the *next state*.

A transition matrix P for a Markov chain with n states can be represented as follows:

$$P = \begin{array}{c|cccc} & & X_{t+1} & & \\ X_t & 1 & 2 & \cdots & n \\ \hline 1 & p_{11} & p_{12} & \cdots & p_{1n} \\ 2 & p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n & p_{n1} & p_{n2} & \cdots & p_{nn} \end{array}$$

where the rows represent X_t (current state at time t) and the columns represent X_{t+1} (next state at time $t+1$).

The entry P_{ij} represents the probability of transitioning from state i at time t to state j at time $t+1$. In probability notation, we can express this as:

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

For the weather example, the transition matrix is

$$P = \begin{array}{lcc} & \text{Sunny} & \text{Rainy} \\ \text{Sunny} & 0.8 & 0.2 \\ \text{Rainy} & 0.4 & 0.6 \end{array}$$

The entry in row i and column j gives the probability of moving from state i to state j in a single time step.

Exercise 2 Frank's Hot Dog stand, returns

Working Space

Suppose Frank's hot dog stand is still running strong, but the customers are getting harder to please. The states still remain $\mathcal{S} = \{N, R, I\}$. The transition probabilities have changed, and the new transition matrix is given by:

$$P = \begin{bmatrix} 0.50 & 0.30 & 0.20 \\ 0.15 & 0.65 & 0.20 \\ 0.10 & 0.15 & 0.75 \end{bmatrix}$$

1. Find the probability that a regular customer becomes an inactive customer on their next visit.
2. Find the probability that an inactive customer becomes a regular customer after two visits.
3. Find the probability that a new customer becomes an inactive customer after three visits.

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From this exercise, we saw how to calculate the probability of transitioning from one state to another after multiple steps. This can be done by multiplying the transition probabilities along the path of states and summing over all possible paths.

We can also do this process using matrix multiplication. Specifically, if we want to find the probability of transitioning from one state to another after k steps, we can compute the k-th power of the transition matrix P^k. The entry in row i and column j of P^k will give us the probability of transitioning from state i to state j in exactly k steps.

The transition matrix P gives probabilities for one state change. To find probabilities after two state changes, we compute P^2. This is done by multiplying the transition matrix by itself:

$$P^2 = P \times P$$

For example, using the hot dog stand transition matrix, we would represent the probability of transitioning from a new customer to an regular customer after two steps as:

$$\sum_k P_{N,k} \cdot P_{k,R} = P^2_{N,R}$$

where k ranges over all possible intermediate states (new, regular, inactive).

Therefore, if we want the probability of moving from state i to state j after two steps, we look at the (i, j) entry of P^2, written $(P^2)_{ij}$.

Generally we can represent this as the following equation:

$$P^k_{ij} = P(X_{n+k} = j \mid X_n = i)$$

where P^k_{ij} is the entry in row i and column j of the matrix P^k. This gives us the probability of transitioning from state i to state j in exactly k steps.

$$\text{Probability Distribution after } k \text{ steps} = P^k$$

Note that this is only possible square matrices, which means the number of states must be the same for the current and next state. This is a common assumption in Markov chains, as it allows us to model processes where the set of possible states does not change over time.

Exercise 3 Chemical Change

A chemical changes colors approximately every 30 minutes. The state space of this chemical is $\mathcal{S} = \{\text{Green}, \text{Blue}\}$.

$$P = \begin{bmatrix} 0.2 & 0.8 \\ 0.1 & 0.9 \end{bmatrix}$$

Find the P^3, the transition matrix after 90 minutes. Then, find $P(X_{n+3} = \text{Green} \mid X_n = \text{Blue})$.

Working Space

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1.3 Steady State Distribution

For many Markov chains, these probabilities eventually stabilize (recall **Law of Large Numbers**, P^k as $k \to \infty$) as we request more and more transitions or future states (recall Law of Large Numbers). That is, after a large number of transitions, the proportion of the probability of each state approaches a fixed value. This long-run behavior is known as the *steady-state distribution* or *equilibrium distribution* of the Markov chain.

A steady-state distribution has a special property: applying the transition matrix does not change it. If $\mathbf{x}$ is a steady-state vector and P is the transition matrix, then

$$P\mathbf{x} = \mathbf{x}.$$

In other words, once the chain reaches equilibrium, the probabilities remain unchanged from one time step to the next. This equation is also the eigenvector equation with eigenvalue 1. Therefore, steady-state distributions can be found by determining the eigenvector corresponding to the eigenvalue 1 and then normalizing its entries so that they sum to 1.

The following example illustrates how eigenvectors can be used to determine the long-run proportion of time a Markov chain spends in each state.

Exercise 4 Markov Chain in the form of Transition Matrix

Working Space

This question is taken from an IB practice exam. A Markov Chain has two possible states, A and B. The transition matrix P is given by:

$$P = \begin{bmatrix} p & 1-q \\ 1-p & q \end{bmatrix}$$

Given one of the eigenvalues is 1.

1. Solve for the eigenvector corresponding to the eigenvalue 1.
2. Find an expression for the proportion of time the Markov chain spends in state A in the long run in terms of p and q.
3. Hence or otherwise, find the proportion of time spent in state A when $p = 0.3$ and $q = 0.6$.

Answer on Page 17

1.4 Applications

1.4.1 PageRank

Imagine an individual browsing the internet. They start on a random webpage and click on links to navigate to other pages. The probability of moving from one webpage to another can be modeled as a Markov chain, where each webpage represents a state and the links between pages represent transitions with certain probabilities.

In this case, each page is a *state* and each link from one page to another represents a *transition* with a certain probability. The transition probabilities can be determined based on the structure of the web, such as the number of links on each page and the likelihood of a user clicking on those links. If a page has more links, the probability of transitioning to any particular linked page is lower. For example, if page i has 3 *outgoing links*, the probability of transitioning to each linked page would be $\frac{1}{3}$. The more links to a given page that exist (called *incoming links*), the more "important" that webpage is. Since these links are directed and each form a Markov Chain, recognize that this is *directed graph*.

Now, if we want to find the long-run proportion of time a user spends on each webpage, we can use the *steady-state distribution* of this Markov chain. The steady-state distribution will give us the PageRank of each webpage, which is a measure of its importance based on the structure of the web. A few issues arise in this model, such as the presence of *dead ends* (pages with no outgoing links) and *spider traps* (sets of pages that link to each other but not to the rest of the web). To find the ultimate page rank, we can use a modified version of the transition matrix that consists of a stabilized probability

$$\pi_{n+1} = \pi_n P$$

eventually this will converge to a steady-state distribution π such that $\pi P = \pi$. This is the PageRank vector (similar to the eigenvector discussed above), which gives the relative importance of each webpage in the long run.

There may be an additional *damping factor* (d) that is introduced to compensate for the possibility that a user randomly stops following links and instead jumps to a different random page. This is often set to around 0.85. The probability that they instead jump to any random page is $1-d$, usually 0.15. In matrix terms, this leads to a modified transition model

$$G = dP + (1-d)\frac{1}{n}J,$$

where J is the all-ones matrix and n is the number of pages. This ensures the resulting chain is irreducible and aperiodic, so the steady-state distribution exists and is unique.

1.4.2 Hidden Markov Models and NLP

Hidden Markov Models (HMMs) are an expanded version of a Markov Process. In addition to having probabilities about transition from state to state, it also has influencing *observations*, called *emissions*. An emission is a *visible* output generated by those the states in play, often which are hidden or not visible.

Assume you have a patient in a hospital with two possible states: $\mathcal{S} = \{\text{Sick}, \text{Healthy}\}$. You cannot know the exact state at a current time with certainty. However, you have three observable factors: $\mathcal{O} = \text{no fever}, \text{fever}, \text{cough}$. You may also see the emission states symbolized with $\mathcal{E}$.

For each state change observation (for this example, lets say once a day), you need both the transition matrix and the *emission matrix*. We will define the transition matrix T and emission matrix E as follows:

$$S = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}, \qquad E = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.7 & 0.1 & 0.2 \end{bmatrix}$$

Note that the emission matrix is not necessarily square. In real life examples, there would be probably hundreds of possible states, each with tons of possible emissions.

The emission matrix describes the probability of observing each symptom given a particular hidden state. For example,

$$P(\text{fever} \mid \text{Sick}) = 0.5,$$

while

$$P(\text{no fever} \mid \text{Healthy}) = 0.7.$$

Suppose that over three consecutive days we observe the symptom sequence

$$\text{fever} \to \text{cough} \to \text{no fever}.$$

These observations alone do not reveal the patient's true health state. One possible hidden state sequence is

$$\text{Sick} \to \text{Sick} \to \text{Healthy},$$

while another is

$$\text{Healthy} \to \text{Sick} \to \text{Healthy}.$$

The goal of an HMM is to infer the most likely hidden state sequence that generated the observed emissions.

While we won't discuss them in this book, there are algorithms such as the Forward and Viterbi algorithms that can track and guess that most likely future state given the states available, or most likely emissions given a set of states. These use *dynamic programming*, a method that programming similar to excel processing.

1.4.3 Natural Language Processing

Another strong use of HMM is the basics of Natural Language Processing, and ultimately, predictive AI text, such as recommended words above your phone's keyboard.

Say we have a set of states represent parts of speech

$$\mathcal{S} = \{\text{Noun}, \text{Verb}, \text{Adjective}, \dots\}$$

These states are hidden because when you read a sentence, you only see the words, not their correct grammatical labels.

The observations are the words themselves:

$$\mathcal{O} = \{\text{dog}, \text{runs}, \text{fast}, \dots\}$$

Suppose we are given the sentence `dog runs fast`. The observations are known, but the hidden states are not. One possible hidden-state sequence is

$$\texttt{Noun} \rightarrow \texttt{Verb} \rightarrow \texttt{Adverb}$$

The transition matrix models how likely one grammatical tag is to follow another. For example,

$$P(\text{Verb} \mid \text{Noun}) = 0.6$$

might represent that verbs frequently follow nouns in the natural spoken language. The emission matrix models how likely a word is to be generated by a particular tag. For example,

$$P(\texttt{dog}|\text{Noun}) = 0.4, P(\texttt{runs}|\text{Verb}) = 0.5$$

Exercise 5 NLP Exercise

Working Space

Suppose an HMM uses the following part-of-speech tags:

$$\mathcal{S} = \{\text{Noun (N)}, \text{Verb (V)}, \text{Adjective (A)}\}.$$

From a training dataset of words, the following transition probabilities were estimated:

$$T = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.4 & 0.2 \\ 0.6 & 0.2 & 0.2 \end{bmatrix}$$

where the rows and columns correspond to the states

$$(\text{Noun}, \text{Verb}, \text{Adjective}).$$

Find the probability that an adjective occurs after a noun followed by a verb. That is, compute

$$P(N \to V \to A).$$

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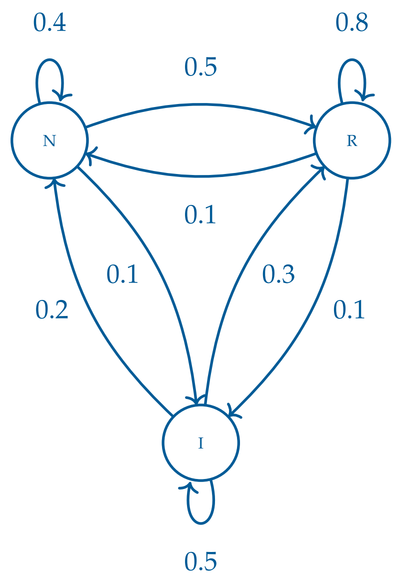
Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 3)

1. The state space of this Markov chain is $\mathcal{S} = \{N, R, I\}$ where N represents new customers, R represents regular customers, and I represents inactive customers.

2. 

3. $$P(X_{n+1} = R \mid X_n = I, X_{n-1} = R, X_{n-2} = N) = P(X_{n+1} = R \mid X_n = I)$$
Probability: 30%

Answer to Exercise 2 (on page 6)

1. Regular $\to$ Inactive: $P(X_{n+1} = I \mid X_n = R) = 0.20$.
2. Inactive $\to$ Regular after two visits: We need to find $P(X_{n+2} = R \mid X_n = I)$. This can be calculated by two multiplications and summing over all possible intermediate states:
   - $I \to N \to R = 0.10 \times 0.30 = 0.03$

- $I \to R \to R = 0.15 \times 0.65 = 0.0975$
- $I \to I \to R = 0.75 \times 0.15 = 0.1125$

$$0.03 + 0.0975 + 0.1125 = 0.24$$

3. New $\to$ Inactive after three visits: We need to find $P(X_{n+3} = I \mid X_n = N)$. This can be calculated by *three* multiplications and summing over all possible intermediate states:

   - $N \to N \to N \to I = 0.50 \times 0.50 \times 0.20 = 0.05$
   - $N \to N \to R \to I = 0.50 \times 0.30 \times 0.20 = 0.03$
   - $N \to N \to I \to I = 0.50 \times 0.20 \times 0.75 = 0.075$
   - $N \to R \to N \to I = 0.30 \times 0.15 \times 0.20 = 0.009$
   - $N \to R \to R \to I = 0.30 \times 0.65 \times 0.20 = 0.039$
   - $N \to R \to I \to I = 0.30 \times 0.20 \times 0.75 = 0.045$
   - $N \to I \to N \to I = 0.20 \times 0.10 \times 0.20 = 0.004$
   - $N \to I \to R \to I = 0.20 \times 0.15 \times 0.20 = 0.006$
   - $N \to I \to I \to I = 0.20 \times 0.75 \times 0.75 = 0.1125$

$$0.050 + 0.030 + 0.075 + 0.009 + 0.039 + 0.045 + 0.004 + 0.006 + 0.1125 = 0.3705$$

Answer to Exercise 3 (on page 7)

To find P^3, we need to compute the matrix multiplication of P with itself three times. First, we compute P^2:

$$P^2 = \begin{bmatrix} 0.2(0.2) + 0.8(0.1) & 0.2(0.8) + 0.8(0.9) \\ 0.1(0.2) + 0.9(0.1) & 0.1(0.8) + 0.9(0.9) \end{bmatrix} = \begin{bmatrix} 0.12 & 0.88 \\ 0.11 & 0.89 \end{bmatrix}$$

Then find P^3 from $P^2 \cdot P$

$$\begin{aligned} P^3 &= \begin{bmatrix} 0.12 & 0.88 \\ 0.11 & 0.89 \end{bmatrix} \begin{bmatrix} 0.2 & 0.8 \\ 0.1 & 0.9 \end{bmatrix} \\ &= \begin{bmatrix} 0.12(0.2) + 0.88(0.1) & 0.12(0.8) + 0.88(0.9) \\ 0.11(0.2) + 0.89(0.1) & 0.11(0.8) + 0.89(0.9) \end{bmatrix} \\ &= \begin{bmatrix} 0.112 & 0.888 \\ 0.111 & 0.889 \end{bmatrix} \end{aligned}$$

So

$$P^3_{\text{Blue,Green}} = P(X_{n+3} = \text{Green} \mid X_n = \text{Blue}) = 0.111$$

, which means that if the chemical is currently blue, there is an 11.1% chance it will be green after 90 minutes.

Answer to Exercise 4 (on page 9)

1. We want to solve for $P\mathbf{x} = \mathbf{x}$. Let $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$. Then we have

$$\begin{bmatrix} p & 1-q \\ 1-p & q \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

   Then this gives us the following system of equations:

$$\begin{aligned} px + (1-q)y &= x \\ (1-p)x + qy &= y \end{aligned}$$

$$x = 1-q, \qquad y = 1-p$$

   Therefore, the eigenvector corresponding to the eigenvalue 1 is

$$\begin{bmatrix} 1-q \\ 1-p \end{bmatrix}$$

2. Normalizing it gives us $(1-q) + (1-p) = 2-p-q$. Therefore, the steady-state distribution is

$$\begin{bmatrix} \frac{1-q}{2-p-q} \\ \frac{1-p}{2-p-q} \end{bmatrix}$$

   Which for A is $\frac{1-q}{2-p-q}$.

3. Plugging in $p = 0.3$ and $q = 0.6$ gives us $\frac{1-0.6}{2-0.3-0.6} = \frac{0.4}{1.1} = \frac{4}{11} = 0.364$ of the proportion of time spent in state A.

Answer to Exercise 5 (on page 13)

By the Markov property,

$$P(N \to V \to A) = P(V \mid N)\, P(A \mid V).$$

Substituting the values from the transition matrix,

$$P(N \to V \to A) = (0.5)(0.2) = 0.10.$$

Therefore,

$$\boxed{P(N \to V \to A) = 0.10.}$$

Thus, according to the model, the probability of observing the tag sequence

$$\text{Noun} \to \text{Verb} \to \text{Adjective}$$

is 10%.

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